

Eros Fani`

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Professional and teaching experience

Machine Learning Researcher

Oct 2025 – Jul 2026

GENSYN

- Led research on **model merging**, **mixture of experts**, **LLMs routing**, and **decentralized ML**
- Designed and implemented a **decentralized AI inference platform** that orchestrates multiple remote LLM providers behind a unified conversational interface (public release forthcoming)

Postdoctoral Fellow & Software Engineer

Nov 2024 – Oct 2025

BASQUE CENTER FOR APPLIED MATHEMATICS (BCAM)

- Lead developer, **Act.AI project** (PI: Prof. N. Quadrianto)

MSc thesis supervisor

Sep 2023 - Oct 2024

POLYTECHNIC UNIVERSITY OF TURIN (POLITO)

- Topic: Clustered Federated Learning
- Resulted in a publication at **ICML25**

Teaching Assistant

Sep 2022 - Sep 2023

POLYTECHNIC UNIVERSITY OF TURIN (POLITO)

- Graduate courses: Machine Learning and Deep Learning ◦ Advanced Machine Learning ◦ Data Analytics and Artificial Intelligence ◦ Web Applications I
- Undergraduate course: Computer Science

MSc thesis supervisor

Sep 2022 - Jul 2023

POLYTECHNIC UNIVERSITY OF TURIN (POLITO)

- Topic: Federated Visual Geo-Localization
- Resulted in a publication at **FedVision @ CVPR 2024**

Teaching collaborator as an undergrad

Sep 2018 - Mar 2019

POLYTECHNIC UNIVERSITY OF TURIN (POLITO)

- Undergraduate courses: Algorithms and Programming ◦ Mathematical Analysis I ◦ Mathematical Analysis II

Education

Ph.D. in Computer and Control Engineering

Nov 2021 - Jul 2025

POLYTECHNIC UNIVERSITY OF TURIN (POLITO)

- Thesis title: “**Addressing Heterogeneity in Federated Learning for Real-world Vision Applications**”
- Advisor: Prof. B. Caputo · Polytechnic University of Turin (PoliTo)

ELLIS Ph.D. program - Academic track

Nov 2021 - Jul 2025

EUROPEAN LABORATORY FOR LEARNING AND INTELLIGENT SYSTEMS (ELLIS)

- Primary Host: Prof. B. Caputo · Polytechnic University of Turin (PoliTo)
- Exchange Host: Prof. N. Quadrianto · University of Sussex (UoS)

Visiting period at the University of Sussex for the ELLIS Exchange

May 2024 - Oct 2024

UNIVERSITY OF SUSSEX (UOS), EUROPEAN LABORATORY FOR LEARNING AND INTELLIGENT SYSTEMS (ELLIS)

- Research Advisor: Prof. N. Quadrianto · University of Sussex (UoS)

Visiting period at ELLIS Alicante ELLIS ALICANTE, EUROPEAN LABORATORY FOR LEARNING AND INTELLIGENT SYSTEMS (ELLIS) • Research Advisor: Prof. N. Oliver · ELLIS Alicante	Sep 2024
MSc in Data Science and Engineering POLYTECHNIC UNIVERSITY OF TURIN (POLITO) • Final grade: 110/110 cum laude • Thesis title: On the Challenges of Class Imbalance in Federated Learning for Semantic Segmentation	Oct 2019 - Oct 2021
BSc in Computer Engineering POLYTECHNIC UNIVERSITY OF TURIN (POLITO) • Final grade: 108/110	Oct 2016 - Jul 2019
Young Talent Project POLYTECHNIC UNIVERSITY OF TURIN (POLITO), CRT FOUNDATION • Study program of excellence open only to the 2% top-tier students admitted to the PoliTo for the A.Y. 2016/17	Oct 2016 - Jul 2019
High school diploma LICEO SCIENTIFICO STATALE M. CURIE GIULIANOVA • Final grade: 100/100 cum laude	Sep 2011 - Jul 2016

Additional courses

Warm up your muscles: a ten-stage marathon to become the protagonist of your financial future ↗ MUSEO DEL RISPARMIO, INTESA SAN PAOLO	Feb 2025 - Apr 2025
Fundamentals of Reinforcement Learning ↗ COURSERA, UNIVERSITY OF ALBERTA	Feb 2022
Sample-based Learning Methods ↗ COURSERA, UNIVERSITY OF ALBERTA	Jan 2022
Security Soft Start ↗ ONSTAIRS ACADEMY	Jun 2020

Attended schools and programs

Mediterranean Machine Learning (M2L) Summer School AI EDUCATION FOUNDATION	Thessaloniki, Greece Aug 2023
ELLIS Doctoral Symposium 2021 UNIVERSITY OF TÜBINGEN, MAX-PLANCK INSTITUTE FOR INTELLIGENT SYSTEMS	Tübingen, Germany Oct 2021
Vodafone IoT Academy VODAFONE, POLYTECHNIC UNIVERSITY OF TURIN (POLITO)	Turin, Italy Jan 2020 - Jul 2020
Summer Orientation School Sant'Anna SCUOLA SUPERIORE SANT'ANNA	Volterra, Italy Jun 2015
Summer School for Experimental Sciences GRAN SASSO NATIONAL LABORATORIES (LNGS) OF THE ITALIAN NATIONAL INSTITUTE FOR NUCLEAR PHYSICS (INFN)	Assergi, Italy Jun 2014

Publications

E. Fani , O. Ersoy. “IR3DE: A Linear Router for Large Language Models”. Resource-Adaptive Foundation Model Inference (AdaptFM) workshop of the International Conference on Machine Learning, 2026.	AdaptFM-ICML26
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E. Fani , O. Ersoy. “ <i>Training-Free Dynamic Upcycling of Expert Language Models</i> ”. Scaling Post-training for LLMs (SPOT) workshop of the International Conference on Learning Representations, 2026.	SPOT-ICLR26
• Extended version under review at a major ML conference.	
G. D. Németh, E. Fani , et. al. “ <i>Diversity-Driven Learning: Tackling Spurious Correlations and Data Heterogeneity in Federated Models</i> ”. Third IEEE International Conference on Federated Learning Technologies and Applications, 2025.	FLTA25
E. Fani , et. al. “ <i>Resource-Efficient Personalization in Federated Learning With Closed-Form Classifiers</i> ”. IEEE Access, 2025	IEEE ACCESS
A. Licciardi, ..., E. Fani , et. al. “ <i>Interaction-Aware Gaussian Weighting for Clustered Federated Learning</i> ”. Forty-second International Conference on Machine Learning, 2025.	ICML25
E. Fani , et. al. “ <i>Accelerating Heterogeneous Federated Learning with Closed-form Classifiers</i> ”. Forty-first International Conference on Machine Learning, 2024.	ICML24
M. Dutto, ..., E. Fani , et. al. “ <i>Collaborative Visual Place Recognition through Federated Learning</i> ”. FedVision Workshop in Conjunction with the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024.	FEDVISION CVPR24
E. Fani , et. al. “ <i>Fed3R: Recursive Ridge Regression for Federated Learning with strong pre-trained models</i> ”. International Workshop on Federated Learning in the Age of Foundation Models in Conjunction with NeurIPS, 2023.	FL@FM NEURIPS23
E. Fani , et. al. “ <i>FedDrive v2: an Analysis of the Impact of Label Skewness in Federated Semantic Segmentation for Autonomous Driving</i> ”. 5th Italian Conference on Robotics and Intelligent Machines, 2023.	I-RIM23
D. Shenaj*, E. Fani *, et. al. “ <i>Learning Across Domains and Devices: Style-Driven Source-Free Domain Adaptation in Clustered Federated Learning</i> ”. IEEE/CVF winter conference on applications of computer vision, 2023.	WACV23
L. Fantauzzo*, E. Fani *, et. al. “ <i>FedDrive: Generalizing Federated Learning to Semantic Segmentation in Autonomous Driving</i> ”. IEEE/RSJ International Conference on Intelligent Robots and Systems, 2022.	IROS22

*Equal contribution.

Peer-reviewing

Conference on Neural Information Processing Systems (NeurIPS)	2026, 2025, 2024
Resource-Adaptive Foundation Model Inference workshop of the International Conference on Machine Learning (AdaptFM-ICML)	2026
Scaling Post-training for LLMs workshop of the International Conference on Learning Representations (SPOT-ICLR)	2026
IEEE Robotics and Automation Letters (RA-L)	2025, 2022
European Conference on Artificial Intelligence (ECAI)	2025
• Invited as Program Committee Member	
IEEE/CVF Computer Vision and Pattern Recognition Conference (CVPR)	2025
Transactions on Machine Learning Research (TMLR)	2024
European Conference on Computer Vision (ECCV)	2024
• Awarded as Outstanding Reviewer	
IEEE International Conference on Development and Learning (ICDL)	2023
European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML-PKDD)	2023
IEEE International Conference on Robotics and Automation (ICRA)	2023
Learning from Limited and Imperfect Data workshop of the European Conference on Computer Vision (L2ID-ECCV)	2022

Invited talks

Speaker in a Gensyn X Space [↗](#)

Jun 2026

Presenting the research paper: IR3DE: A LINEAR ROUTER FOR LARGE LANGUAGE MODELS

- Engaging in live Q&A with audience

Speaker in a Gensyn X Space [↗](#)

May 2026

Presenting the research paper: TRAINING-FREE DYNAMIC UPCYCLING OF EXPERT LANGUAGE MODELS

- Engaging in live Q&A with audience

Video presentation at the Flower AI summit, organized by Flower AI [↗](#)

Apr 2026

Presenting the research paper: TRAINING-FREE DYNAMIC UPCYCLING OF EXPERT LANGUAGE MODELS

Honors, awards and achievements

Outstanding Reviewer award at ECCV24

Sep 2024

EUROPEAN CONFERENCE ON COMPUTER VISION 2024, EUROPEAN COMPUTER VISION ASSOCIATION (ECVA)

Scholarship award for the participation to the Mediterranean Machine Learning (M2L) Summer School

Aug 2023

AI EDUCATION FOUNDATION

- Selected as one of the **top 10 students** among 805 applicants (top 1.25%) and 180 participants (top 5.55%)

Granted the WACV23 registration fee waiver

Nov 2022

IEEE/CVF WINTER CONFERENCE ON APPLICATIONS OF COMPUTER VISION 2023

Two-times RoboMensa National Winner

2015 - 2016

MENSA ITALIA – THE HIGH I.Q. SOCIETY

- Logic Contest - **1st place** in both Individual and Teams categories

Two-times National Finalist at the Italian Mathematics Olympic Games in Teams

2015 - 2016

ITALIAN MATHEMATICAL UNION

Three podiums at the RomeCup Rescue Secondary National Finals

2014 - 2016

DIGITAL WORLD FOUNDATION

- **1st place** (2016) – **2nd place** (2014, 2015)

National Finalist at the Italian Informatics Olympic Games

2015

MINISTRY OF INSTRUCTION, UNIVERSITY AND RESEARCH (MIUR)

National Finalist at the Mathematical Games of the Mediterranean

2013

ITALIAN ACADEMY FOR THE PROMOTION OF MATHEMATICS

Technical blog posts

Look Beyond One-Size-Fits-All LLMs with IR3DE [↗](#)

Jun 2026

Blog post on the paper IR3DE: A LINEAR ROUTER FOR LARGE LANGUAGE MODELS

Stop Multitask Training. Just DUME. [↗](#)

Apr 2026

Blog post on the paper TRAINING-FREE DYNAMIC UPCYCLING OF EXPERT LANGUAGE MODELS

Accelerating Heterogeneous Federated Learning with Closed-form Classifiers [↗](#)

May 2024

Blog post on the paper ACCELERATING HETEROGENEOUS FEDERATED LEARNING WITH CLOSED-FORM CLASSIFIERS

FedDrive: a Federated Learning Benchmark for Semantic Segmentation in Autonomous Driving [↗](#)

Sep 2022

Blog post on the paper FEDDRIVE: GENERALIZING FEDERATED LEARNING TO SEMANTIC SEGMENTATION IN AUTONOMOUS DRIVING

Selected media mentions

The Missing Dispatcher for Expert Models [↗](#)

Jun 2026

X article from Gensyn — Coverage of invited X Space discussion on the paper IR3DE: A LINEAR ROUTER FOR LARGE LANGUAGE MODELS

Training-Free Composition of Expert Models [↗](#)

May 2026

X article from Gensyn — Coverage of invited X Space discussion on the paper TRAINING-FREE DYNAMIC UPCYCLING OF EXPERT LANGUAGE MODELS

Languages

Italian - native

English - full professional proficiency

Spanish - elementary proficiency

Coding skills

Python, C, PyTorch, Numpy, Scikit-Learn, WandB, Matplotlib, HTML, HTML5, HTML Bootstrap, CSS, Jax, FedJax, Haiku, Flax, Flower, Textual, Java, C++, Bash, Matlab, Javascript, ReactJS, Vite, PHP, R, Apache Spark, Apache Hadoop